Blueprint v2.1

Section 1 — Philosophical Foundations & Scope

1.1 Research Positioning

This model is not a predictive injury classifier.

It is a **stochastic health capital governance system** designed to:

- Represent latent physiological evolution under workload.
- Quantify injury as a first-hitting-time event in multidimensional space.
- Support structured workload decisions under uncertainty.
- Enforce medically imposed catastrophic risk constraints.
- Produce interpretable performance–risk trade-offs.

The model explicitly integrates:

- Continuous-time physiological evolution
- Discrete-time decision control
- Partial observability
- Approximate stochastic control
- Multi-state injury regimes

1.2 Primary Decision Context

The system is designed for joint use by:

- Coaching staff (performance optimisation)
- Medical staff (risk governance)

Decision structure is hierarchical:

- Medical staff impose catastrophic injury probability constraint.
- Coaching staff choose workload within feasible region.

The model does not represent federation-level scheduling decisions at this stage.

1.3 Season Horizon

Primary horizon:

$$t \in [0, T]$$

where T represents one competitive season.

Career-level sustainability is outside primary scope but structurally extendable.

1.4 Core Design Principles

The architecture satisfies:

1. **Low-dimensional latent state** (tractable, identifiable).
2. **Time-scale separation** (prevent blame spreading).
3. **Hybrid dynamics** (continuous + jump).
4. **Injury defined solely by region entry**.
5. **Piecewise-constant control** (realistic implementation).
6. **Particle-based state estimation** (no linear Gaussian fiction).
7. **Approximate optimisation** (no false analytical optimality).
8. **Explicit irreducible injury component**.

These principles are structural commitments.

Section 2 — Time & Control Architecture

2.1 Continuous Health Evolution

Health evolves in continuous time:

$$t \in [0, T]$$

where T denotes the full competitive season.

Continuous evolution allows:

- Within-game fatigue accumulation
- Intra-day recovery
- Acute shock arrival at arbitrary time
- Travel and circadian effects
- Rehabilitation dynamics

State evolution is governed by stochastic differential equations with jump components.

2.2 Decision Epochs

Workload decisions are not continuously adjusted.

Define decision epochs:

$$0 = t_0 < t_1 < \cdots < t_K = T$$

At each t_k:

- Coaching and medical staff observe updated health proxies.
- Filtering is performed.
- A workload decision is made for the next interval.

Within interval:

$$u_t = u_k \text{for } t \in [t_k, t_{k+1})$$

This creates a **semi-Markov decision structure**.

2.3 Interpretation of Control Intervals

Intervals may correspond to:

- Individual games
- Weekly training blocks
- Recovery microcycles
- Rehabilitation phases

Control is therefore operationally implementable.

2.4 Prevention of Unrealistic Infinitesimal Adjustments

Continuous control would allow:

- Arbitrarily fine "health preservation" micromanagement.
- Unrealistically smooth workload modulation.
- Optimistic exploitation of diffusion structure.

Piecewise-constant control prevents this by:

- Enforcing realistic managerial rigidity.
- Introducing unavoidable exposure between decisions.
- Creating meaningful trade-offs across intervals.

2.5 Information Timing

At each decision epoch t_k:

1. Observations Y_{t_k} are collected.
2. Particle filtering updates belief distribution.
3. Policy selects u_k.
4. Health evolves continuously until t_{k+1}.

Thus control is based on filtered belief, not true state.

2.6 Admissible Control Space

Control must satisfy:

$$u_k \in \mathcal{U}(\mathcal{S}_{t_k})$$

where feasible workload depends on current injury regime:

- Healthy: full admissible range.
- Fatigued: restricted upper bound.
- Minor Injury: rehabilitation range.
- Major Injury: zero.
- Catastrophic: absorbing.

This prevents policy from violating medical constraints structurally.

2.7 Implications for Optimisation

Because:

- State evolves continuously,
- Control changes discretely,

The system becomes:

Continuous-time stochastic dynamics

- Discrete-time decision problem
- Partial observation
- Regime switching

This avoids unrealistic HJB complexity while preserving realism.

Section 3 — State Space Design & Identifiability Architecture

3.1 Latent Health Capital Vector

For each player:

$$\mathbf{H}_t = (S_t, F_t, R_t, D_t)$$

where:

- S_t: Structural Integrity Capital
- F_t: Neuromuscular Fatigue Load
- R_t: Recovery Capacity Index
- D_t: Cumulative Micro-Damage Reserve

State dimension is fixed at 4.

This dimension will not expand.

Additional observables enter only as parameter modifiers, not new states.

3.2 Structural Distinctness of Components

Each component must explain distinct phenomena.

Structural Integrity S_t

- Medium-time-scale variable.
- Directly affected by accumulated load and jumps.
- Governs catastrophic vulnerability.
- Weak short-term performance impact.
- Strong long-term injury relevance.

Fatigue F_t

- Fast-time-scale variable.
- Explains short-term performance drop.
- Explains acute jump intensity increases.
- Highly responsive to workload changes.

Recovery Capacity R_t

- Medium-slow-time-scale variable.

- Modulates rate of fatigue clearance.
- Modulates structural repair.
- Explains persistent slow recovery phenomena.

Micro-Damage D_t

- Slow-time-scale variable.
- Accumulates subclinically.
- Weak short-term performance impact.
- Amplifies jump severity.
- Explains delayed injury events.

3.3 Time-Scale Separation (Critical Identifiability Scaffold)

To prevent the particle filter from reallocating explanatory weight arbitrarily:

We impose structural time-scale hierarchy:

$$\tau_F \ll \tau_S \lesssim \tau_R \ll \tau_D$$

Where:

- τ_F: characteristic fatigue adjustment time (days)
- τ_S: structural decay time (weeks)
- τ_R: recovery adaptation time (weeks)
- τ_D: micro-damage accumulation time (months)

Implementation via:

- Drift magnitude ordering
- Diffusion magnitude ordering
- Bounded curvature constraints

3.4 Diffusion Constraints

Diffusion magnitudes satisfy:

- $\sigma_F$moderate (captures day-to-day fluctuation)
- $\sigma_S$small
- $\sigma_R$small

- $\sigma_D$very small

This prevents:

- Slow states explaining fast performance noise.
- Structural capital mimicking fatigue volatility.

3.5 Drift Curvature Constraints

To avoid confounding:

- Fatigue has strong mean-reversion.
- Structural capital has weak mean-reversion.
- Micro-damage has no spontaneous decay without recovery input.
- Recovery capacity changes slowly under chronic stress.

These curvature constraints are structural priors, not empirical guesses.

3.6 Cross-Interaction Restrictions

Cross-effects are allowed but constrained:

- Fatigue accelerates structural decay.
- Micro-damage reduces effective structural resilience.
- Low recovery slows fatigue clearance.
- Chronic fatigue reduces recovery capacity.

But:

- Structural integrity does not directly drive fatigue.
- Micro-damage does not directly affect short-term performance.

This avoids circular explanations.

3.7 Identifiability Strategy

Identifiability is enforced through:

1. Time-scale separation.
2. Diffusion magnitude ordering.
3. Restricted cross-interaction graph.
4. Partial parameter sharing across players.
5. Observability mapping design (later section).

The model does not attempt to identify fully unconstrained nonlinear dynamics.

Structural discipline prevents overfitting.

Section 4 — Explicit Health Dynamics

We now formalise the continuous-time latent health process:

$$\mathbf{H}_t = (S_t, F_t, R_t, D_t)$$

The system evolves as a **nonlinear jump–diffusion process**:

$$d\mathbf{H}_t = \mathbf{b}(\mathbf{H}_t, u_k, Z_t)\, dt + \Sigma\, dW_t + \mathbf{J}_t\, dN_t$$

where:

- u_k is piecewise-constant control
- Z_t are observed covariates
- W_t is 4D Brownian motion
- N_t is state-dependent Poisson process
- $\mathbf{J}_t$ is jump magnitude vector

We now define each component explicitly.

4.1 Fatigue Dynamics F_t (Fast Time Scale)

$$dF_t = [\alpha_F u_k - \beta_F R_t F_t - \kappa_F F_t]dt + \sigma_F dW_t^{(F)}$$

Where:

- $\alpha_F > 0$ workload accumulation rate.
- $\beta_F > 0$ recovery-modulated fatigue clearance.
- $\kappa_F > 0$ intrinsic mean-reversion.
- σ_F moderate diffusion.

Time-scale:

$$\tau_F \sim \frac{1}{\kappa_F}$$

This is short (days).

Fatigue is the primary fast-moving variable explaining short-term performance variation.

4.2 Structural Integrity S_t(Medium Time Scale)

$$dS_t = [-\alpha_S u_k(1 + \gamma_F F_t) - \gamma_D D_t + \beta_S R_t]dt + \sigma_S dW_t^{(S)} + J_t^{(S)} dN_t$$

Interpretation:

- Workload depletes structure.
- Fatigue amplifies structural wear.
- Micro-damage accelerates structural degradation.
- Recovery capacity repairs structure.
- Jump causes discrete structural loss.

Constraints:

$$\sigma_S \ll \sigma_F$$

Structural capital cannot mimic fatigue volatility.

4.3 Recovery Capacity R_t(Medium-Slow Time Scale)

$$dR_t = [-\alpha_R u_k - \eta_F F_t + \beta_R Z_t^{(rec)} - \kappa_R(R_t - \bar{R})]dt + \sigma_R dW_t^{(R)}$$

Where:

- α_R: chronic overload suppression.
- η_F: burnout effect from fatigue.
- $Z_t^{(rec)}$: sleep/nutrition/recovery inputs.
- κ_R: slow mean reversion to baseline $\bar{R}$.
- $\sigma_R$small.

Time scale slower than fatigue.

Recovery capacity cannot fluctuate rapidly.

4.4 Micro-Damage D_t(Slow Time Scale)

$$dD_t = [\alpha_D u_k + \eta_D F_t - \beta_D R_t]dt + \sigma_D dW_t^{(D)} + J_t^{(D)} dN_t$$

Properties:

- Accumulates slowly.

- Cleared gradually via recovery.
- Amplifies jump severity.
- Very small diffusion:

$$\sigma_D \ll \sigma_S$$

Time scale months.

4.5 Jump Process (Shock Events Only)

Jump arrivals:

$$\lambda_t = \lambda_0 + \lambda_F F_t + \lambda_S (S_{crit} - S_t)^+ + \lambda_u u_k + \lambda_Z Z_t^{(context)}$$

Where:

- λ_0: irreducible injury exposure.
- Fatigue increases jump arrival.
- Poor structure increases jump likelihood.
- Workload increases exposure risk.

Important:

Jump occurrence ≠ injury.

Jump only perturbs state.

Jump intensity is constrained to remain non-negative and Lipschitz continuous in state variables to ensure well-defined stochastic dynamics.

4.6 Jump Magnitude Distribution

Jump vector:

$$\mathbf{J}_t = (J_t^{(S)}, 0, 0, J_t^{(D)})$$

Magnitude depends on state:

$$\mathbb{E}[J_t^{(S)}] = -\phi_1 (1 - S_t) - \phi_2 D_t$$

Higher micro-damage → larger structural loss.

Fatigue increases variance of jump magnitude.

This captures:

- Healthy player survives shock.
- Fragile player experiences severe loss.

4.7 Boundary Conditions

Each state bounded:

- $S_t \in [0, S_{max}]$
- $F_t \geq 0$
- $R_t \geq 0$
- $D_t \geq 0$

Recovery slows near upper physiological limits.

Structural decay accelerates near lower bound.

Prevents explosive trajectories.

State bounds are enforced either via reflecting boundaries or via drift terms that diverge near physiological limits, ensuring non-explosive trajectories.

4.8 Structural Separation Summary

Fast:

- F_t

Medium:

- S_t

Medium-Slow:

- R_t

Slow:

- D_t

Diffusion ordering:

$$\sigma_F > \sigma_S > \sigma_R > \sigma_D$$

Drift magnitude ordering enforced similarly.

This ensures identifiability.

4.9 Minimum Diffusion Floor

A minimal diffusion floor $\sigma_{min} > 0$ is imposed on all state components, including slow states, to prevent filter collapse and maintain particle diversity.

Interpretation of Diffusion Floor.
The minimal diffusion constraint is not introduced solely for numerical stabilisation. It is interpreted as representing unmodelled micro-physiological variability and measurement–process mismatch that cannot be explicitly captured within the four-dimensional latent structure. Even slow-moving components such as structural integrity and cumulative micro-damage are subject to biological stochasticity arising from cellular repair variability, inflammatory fluctuations, and external perturbations. The diffusion floor therefore preserves both computational stability and physiological realism.

Section 5 — Injury Region Geometry & Regime Switching

5.1 Injury Defined by Region Entry Only

Injury is defined as first entry of the latent state into predefined subsets of health space.

Let:

$$\mathcal{R}_{minor}, \mathcal{R}_{major}, \mathcal{R}_{cat} \subset \mathbb{R}^4$$

be disjoint regions.

Define hitting times:

$$\tau_j = \inf\{t \geq 0 : \mathbf{H}_t \in \mathcal{R}_j\}$$

Injury is **not defined by jump occurrence**.

A jump may:

- Leave the state in healthy region.
- Move state into injury region.
- Push state across catastrophic boundary.

Injury classification depends solely on region geometry.

5.2 Region Geometry

Regions are defined as nonlinear surfaces in 4D space.

Examples (conceptual form only):

Minor Injury region may satisfy:

$$S_t < S_{minor} \text{OR} F_t > F_{critical}$$

Major Injury region may satisfy:

$$S_t < S_{major} \text{AND} D_t > D_{threshold}$$

Catastrophic region may satisfy:

$$S_t < S_{cat}$$

or

$$S_t + \omega D_t < C_{cat}$$

Actual functional forms will be calibrated empirically.

Important structural principle:

Regions depend on joint conditions in 4D space.

They are not scalar risk scores.

5.3 Parametric Geometry Restrictions

5.3.1 Low-Dimensional Parametric Constraint.

Injury regions must be defined using low-dimensional parametric surfaces to preserve identifiability and prevent geometric overfitting.

Each injury region boundary may contain at most three free parameters beyond scaling constants.

Admissible functional forms are restricted to:

- Linear threshold surfaces (e.g., $a_1 S_t + a_2 D_t \leq c$),
- Quadratic forms with diagonal curvature only, or
- Logistic-type threshold functions with fixed slope parameters.

Fully flexible nonlinear 4-dimensional surfaces are explicitly prohibited.

5.3.2 Hierarchical Anchoring Requirement.

The catastrophic injury region must contain an explicit structural integrity component. Specifically, any admissible catastrophic boundary must include either:

$$S_t \leq S_{cat}$$

or a joint structural form such as:

$$S_t + \omega D_t \leq C_{cat}$$

Catastrophic injury cannot be defined independently of structural degradation.

5.3.3 Nesting Enforcement.
Region parameters must be selected such that:

$$\mathcal{R}_{cat} \subset \mathcal{R}_{major} \subset \mathcal{R}_{minor}$$

This nesting constraint is enforced explicitly during calibration.

5.4 Regime Variable

Define regime process:

$$\mathcal{S}_t \in \{Healthy, Fatigued, Minor, Major, Catastrophic\}$$

Regime is deterministic function of health state:

$$\mathcal{S}_t = f(\mathbf{H}_t)$$

No separate stochastic switching process.

5.5 Regime-Dependent Dynamics

Dynamics are modified by regime.

Healthy

Full admissible workload.
Standard drift equations.

Fatigued

Upper bound on workload reduced:

$$u_k \leq u_{fatigue}$$

Performance multiplier penalised.

Drift remains continuous.

Minor Injury

Control restricted to rehabilitation range:

$$u_k \leq u_{rehab}$$

Structural repair term increased.
Jump intensity temporarily elevated (re-injury risk).

Exit occurs when state leaves minor region.

Major Injury

Workload forced to zero:

$$u_k = 0$$

Structural repair dominates drift.
No performance contribution.

Return requires re-entry into non-major region.

Catastrophic Injury

Absorbing state:

If:

$$\tau_{cat} \le T$$

Then:

- Player exits system.
- Performance = 0 for remainder of season.
- No further control decisions.

5.6 Catastrophic Ruin Time

$$\tau_{cat} = \inf\{t \in [0,T]: \mathbf{H}_t \in \mathcal{R}_{cat}\}$$

Season survival probability:

$$\Psi(T) = \mathbb{P}(\tau_{cat} > T)$$

Medical constraint:

$$\Psi(T) \ge 1 - \alpha$$

5.7 Clean Separation from Jump Process

Important structural rule:

Jump occurrence:

- Alters $\mathbf{H}_t$.

Injury:

- Occurs only if post-jump (or continuous drift) state enters region.

Thus:

Jump ≠ injury.
Drift ≠ injury.
Only region crossing defines injury.

This removes overlap.

Section 6 — Performance Functional Architecture

6.1 Instantaneous Performance Contribution

Instantaneous performance at time t:

$$P_t = P(\mathbf{H}_t, u_k, \mathcal{S}_t)$$

Performance is defined only during:

$$\mathcal{S}_t \notin \{\text{Major,Catastrophic}\}$$

Otherwise:

$$P_t = 0$$

6.2 Structural Decomposition

We decompose performance into:

$$P_t = f(u_k) \cdot M(\mathbf{H}_t) + \varepsilon_t^{(P)}$$

Where:

- $f(u_k)$: workload productivity curve
- $M(\mathbf{H}_t)$: health efficiency multiplier
- $\varepsilon_t^{(P)}$: residual match-level randomness

This decomposition prevents conflating workload and health effects.

6.3 Workload Productivity Function $f(u_k)$

Properties:

1. Positive marginal productivity:

$$f'(u_k) > 0$$

2. Diminishing returns:

$$f''(u_k) < 0$$

3. Upper saturation bound.

This ensures:

- Overplaying does not scale performance linearly.
- Optimal interior solutions exist.

6.4 Health Efficiency Multiplier $M(\mathbf{H}_t)$

Define:

$$M(\mathbf{H}_t) = \exp(\quad -a_F F_t - a_D D_t) \cdot (1 + b_R R_t) \cdot (1 - c_S (S_{crit} - S_t)^+)$$

Interpretation:

- Fatigue reduces efficiency.
- Micro-damage reduces efficiency.
- Recovery capacity enhances sustainable output.
- Low structural integrity increases instability penalty.

Important:

Structural integrity affects volatility more than mean (see below).

6.5 Performance Volatility Structure

Performance variance depends on health:

$$\mathrm{Var}(P_t) = \sigma_P^2 + \eta_S (S_{crit} - S_t)^+ + \eta_F F_t$$

Thus:

- Low structure increases performance variance.

- High fatigue increases variability.

This connects naturally to risk-averse objective.

6.6 Regime Penalties

If $\mathcal{S}_t =$ Minor:

$$P_t = \delta_{minor} \cdot f(u_k) M(\mathbf{H}_t)$$

If $\mathcal{S}_t =$ Major:

$$P_t = 0$$

If $\mathcal{S}_t =$ Catastrophic:

$$P_t = 0 \forall t > \tau_{cat}$$

6.7 Seasonal Performance Functional

Total seasonal performance:

$$\Pi = \int_0^T P_t \, dt$$

Because control is piecewise-constant:

$$\Pi = \sum_{k=0}^{K-1} \int_{t_k}^{t_{k+1}} f(u_k) M(\mathbf{H}_t) dt$$

6.8 Risk-Adjusted Coaching Objective

Coach maximises:

$$J(u) = \mathbb{E}[\Pi] - \theta \, \text{Var}(\Pi)$$

Where:

- $\theta \geq 0$measures performance risk aversion.
- $\theta = 0$reduces to risk-neutral optimisation.

Medical constraint remains separate:

$$\mathbb{P}(\tau_{cat} \leq T) \leq \alpha$$

This prevents double-counting injury risk.

The variance term reflects performance volatility only and does not substitute for the catastrophic injury constraint, which remains separately binding.

6.9 Structural Cleanliness Check

Performance depends strongly on:

- Fatigue (fast scale)
- Micro-damage (slow scale)
- Recovery capacity (medium)
- Workload

Structural integrity affects:

- Jump severity
- Performance volatility
- Catastrophic threshold

But not directly large short-term mean shifts.

This preserves identifiability separation.

Section 7 — Information Structure & Particle Filtering Layer

7.0 Filtering Stability Safeguards

- Effective Sample Size (ESS) monitored.
- Rejuvenation kernels applied each decision epoch.
- Tempered likelihood used if needed.

Posterior Contraction Monitoring

Slow-State Posterior Variance Monitoring.
In addition to Effective Sample Size (ESS) monitoring, posterior dispersion of slow-moving states $S_t$and $D_t$is tracked explicitly.

Let:

$$V_{S,k} = \mathrm{Var}(S_{t_k} \mid \mathcal{F}_{t_k}), V_{D,k} = \mathrm{Var}(D_{t_k} \mid \mathcal{F}_{t_k})$$

If posterior variance in either slow state contracts at a rate inconsistent with its characteristic drift time scale (i.e., collapses faster than physically plausible repair or degradation dynamics allow), artificial variance inflation or additional rejuvenation is applied.

This safeguard prevents silent overconfidence in structural integrity or micro-damage estimation, particularly in low-observation or rare-event regimes.

Conservative Override Trigger.
If posterior contraction persists beyond tolerance thresholds, catastrophic risk estimates are adjusted conservatively by widening survival probability uncertainty bands before workload decisions are executed.

7.1 Partial Observability

The true health state:

$$\mathbf{H}_t = (S_t, F_t, R_t, D_t)$$

is latent and unobservable.

At decision epochs t_k, the team observes:

$$Y_{t_k}$$

which may include:

- GPS workload metrics
- Heart rate variability
- Sleep duration
- Biomarkers
- Force plate asymmetry
- Subjective soreness
- Match output statistics
- Injury history

Observations relate to latent state via:

$$Y_{t_k} = h(\mathbf{H}_{t_k}) + \varepsilon_{t_k}$$

where:

- $h(\cdot)$is nonlinear measurement function.
- $\varepsilon_{t_k}$is measurement noise.

7.2 Measurement Design and Identifiability

Measurement functions are structured to reinforce time-scale separation:

- Fatigue proxies load strongly on F_t.
- Structural asymmetry metrics load on S_t.
- Chronic soreness and imaging metrics load on D_t.
- Sleep and hormonal proxies load on R_t.

We do not allow a single observable to load equally on all components.

This prevents observational confounding.

7.3 Belief Representation

At decision epoch t_k, we maintain belief distribution:

$$\pi_k(\mathbf{H}) = p(\mathbf{H}_{t_k} \mid \mathcal{F}_{t_k})$$

where $\mathcal{F}_{t_k}$ contains:

- All past observations
- Past workload decisions
- Known injury regime history

Belief is not approximated as Gaussian.

7.4 Particle Filtering

State estimation uses Sequential Monte Carlo (SMC):

1. Propagate particles under continuous-time dynamics between t_k and t_{k+1}.
2. Apply jump simulation.
3. Update weights using observation likelihood.
4. Resample to prevent degeneracy.

This allows:

- Nonlinear dynamics
- Jump processes
- Non-Gaussian distributions
- Multimodal belief states

7.5 Avoidance of Perfect Bayesian Fiction

We do NOT assume:

- Closed-form filtering.
- Gaussian belief collapse.
- Exact belief-state Markov decision process solution.

Instead:

- Belief is approximated numerically.
- Control decisions use summary statistics (mean, quantiles, ruin probability estimate).
- Policy optimisation uses simulation-based evaluation.

This maintains computational realism.

7.6 Belief Compression for Control

To avoid curse of dimensionality:

Control policy does not depend on full particle cloud.

Instead it may depend on:

- Posterior mean $\widehat{\mathbf{H}}_{t_k}$
- Posterior variance
- Estimated catastrophic ruin probability
- Distance-to-threshold metric

This creates a reduced belief-state representation.

7.7 Estimation Error as Decision Risk

Because health is latent:

- Misestimation can increase ruin probability.
- Overconfidence in structural integrity increases catastrophic risk.
- Underestimating fatigue increases short-term injury risk.

Model outputs explicitly report:

- Confidence intervals on ruin probability.
- Uncertainty bands on structural integrity.

This makes medical governance realistic.

7.8 Information Flow Summary

At each decision epoch:

1. Health evolves continuously.
2. Jumps may occur.
3. Observations collected.
4. Particle filter updates belief.
5. Ruin probability estimated.
6. Coach selects workload for next interval.

This closes the control loop.

Section 8 — Optimisation & Approximate Control Architecture

8.1 Formal Control Problem

At each decision epoch t_k, choose workload u_k to maximise:

$$J(u) = \mathbb{E}[\Pi] - \theta \, \mathrm{Var}(\Pi)$$

subject to:

$$\mathbb{P}(\tau_{cat} \leq T) \leq \alpha$$

and:

$$u_k \in \mathcal{U}(\mathcal{S}_{t_k})$$

State is partially observed via belief π_k.

8.2 True Nature of the Problem

The exact problem is:

- Continuous-time stochastic hybrid system
- With regime switching
- With jump processes
- Under partial observation
- With belief-state dependence

- Over finite horizon

Exact dynamic programming solution is intractable.

We explicitly acknowledge this.

8.3 Policy Representation

Instead of solving HJB exactly, we parameterise policy:

$$u_k = \pi_\theta(\widehat{\mathbf{H}}_{t_k}, \hat{p}_{cat,k}, \mathcal{S}_{t_k})$$

Policy may depend on:

- Posterior mean state
- Estimated catastrophic ruin probability
- Distance-to-threshold
- Injury regime

Policies may be:

- Threshold-based
- Parametric (e.g., sigmoid workload scaling)
- Piecewise linear

This keeps interpretability.

8.4 Certainty-Equivalent Baseline

Baseline approach:

Replace latent state by posterior mean:

$$\mathbf{H}_t \approx \widehat{\mathbf{H}}_t$$

Optimise deterministic approximation locally at each epoch.

This is computationally simple but ignores uncertainty.

Used as benchmark.

8.5 Risk-Aware Approximate Control

Improved policy accounts for uncertainty by:

- Penalising high posterior variance.
- Incorporating ruin probability estimates.

- Using quantile-based state estimates (e.g., worst 10% particle state).

This avoids overconfident exploitation of uncertain health.

8.6 Monte Carlo Policy Evaluation

For each candidate policy:

1. Simulate many season trajectories.
2. Propagate health under jump–diffusion dynamics.
3. Apply particle filtering at decision epochs.
4. Enforce medical constraint.
5. Estimate:
    - Expected performance
    - Performance variance
    - Catastrophic probability

Policy is accepted only if medical constraint satisfied.

8.6.1 Rolling-Horizon Optimisation.

Workload policy is recomputed at each decision epoch $t_k$using a shrinking-horizon framework.

Catastrophic survival constraint is re-evaluated at each decision epoch using updated belief state.

At epoch t_k, optimisation considers the remaining season horizon:

$$[t_k, T]$$

rather than the full original horizon.

Monte Carlo simulation is used to evaluate candidate workload policies over the remaining time interval under current belief state π_k.

The selected workload $u_k$is applied only for interval $[t_k, t_{k+1})$, after which filtering updates and optimisation are repeated.

This rolling-horizon structure:

- Reflects real-world adaptive decision making,
- Prevents early-season overcommitment,
- Naturally increases conservatism near season end,
- Maintains computational feasibility relative to full belief-state dynamic programming.

8.6.2 End-of-Season Behaviour.
As $t_k \to T$, optimisation automatically transitions toward short-horizon behaviour, prioritising immediate performance when catastrophic season-level risk constraint becomes less binding.

8.7 Constrained Optimisation Handling

Medical constraint handled via:

- Hard rejection of policies violating α, or
- Penalty augmentation:

$$J(u) - \Lambda \max(0, \mathbb{P}(\tau_{cat} \leq T) - \alpha)$$

depending on implementation choice.

8.8 Avoidance of Over-Optimism

Continuous dynamics with discrete decisions create:

- Irreducible uncertainty between epochs.
- Jump risk that cannot be eliminated.
- Estimation error that cannot be removed.

This prevents unrealistic "perfect preservation" strategies.

8.9 Governance Interpretation

Outputs include:

- Optimal workload schedule.
- Sensitivity of ruin probability to workload increments.
- Performance–risk frontier.
- Conservative vs aggressive coaching profiles.

Policy is not a black box; it is interpretable.

Section 9 – Risk Metrics & Decision Outputs

9.1 Catastrophic Season Survival Probability

Primary medical metric:

$$\Psi(T) = \mathbb{P}(\tau_{cat} > T)$$

Estimated via Monte Carlo simulation under candidate policy.

Reported as:

- Point estimate
- Confidence interval (due to simulation and filtering uncertainty)

This is the binding medical governance constraint.

9.2 Conditional Short-Horizon Injury Risk

At decision epoch t_k, define:

$$p_{major,k}(\Delta) = \mathbb{P}(\tau_{major} \leq t_k + \Delta \mid \mathcal{F}_{t_k})$$

Used for:

- Weekly workload planning
- Return-to-play decisions
- Back-to-back game assessment

This is a dynamic risk forecast, not a static probability.

9.3 Distance-to-Threshold Metric

Define geometric buffer:

$$\Delta_k = \inf_{\mathbf{h} \in \mathcal{R}_{major}} \| \widehat{\mathbf{H}}_{t_k} - \mathbf{h} \|$$

This measures how close the current belief mean is to injury region.

Also report:

- Worst-particle distance (risk-sensitive version)
- Percentile-based buffer

This gives intuitive "health margin" interpretation.

9.4 Marginal Workload Injury Cost

Define local sensitivity:

$$\frac{\partial}{\partial u_k}\mathbb{P}(\tau_{cat} \leq T)$$

Estimated numerically via small perturbation in workload.

Interpreted as:

Incremental increase in catastrophic probability per unit workload.

This is extremely powerful for:

- Game-by-game minute decisions
- Trade-offs during playoffs
- Managing star players

9.5 Structural Depletion Rate

Expected structural drift:

$$\mathbb{E}[dS_t \mid \mathcal{F}_{t_k}]$$

If persistently negative beyond sustainable threshold:

- Signals long-term deterioration.
- Triggers medical review.

This is a sustainability indicator.

9.6 Fatigue Volatility Indicator

Estimate short-term volatility risk:

$$\mathrm{Var}(F_{t_{k+1}} \mid \mathcal{F}_{t_k})$$

High predicted fatigue volatility implies:

- Increased jump intensity.
- Increased performance instability.

This metric supports conservative play in high-risk windows.

9.7 Performance–Risk Frontier

For a range of admissible policies:

Compute pairs:

$$(\mathbb{E}[\Pi], \Psi(T))$$

Plot efficient frontier:

- Higher performance ↔ lower survival probability.

This provides governance-level visualisation of trade-offs.

9.8 Uncertainty-Aware Reporting

All metrics reported with:

- Monte Carlo confidence intervals.
- Sensitivity bands to parameter uncertainty.
- Posterior dispersion from particle filtering.

This prevents false precision.

9.9 Regime Duration Forecasts

Estimate expected time in:

- Minor injury regime
- Major injury regime
- Healthy state

Under given workload policy.

This helps:

- Roster planning
- Bench depth management
- Rotation forecasting

9.10 Policy Stress Testing

Simulate:

- Increased schedule congestion
- Reduced recovery input
- Elevated baseline jump intensity
- Travel fatigue spikes

Report impact on:

- Survival probability
- Performance expectation
- Structural sustainability

This elevates the model beyond player-level analytics into governance analysis.

Section 10 – Parameter Architecture & Estimation Strategy

10.1 Parameter Categories

Parameters are grouped into seven structured blocks:

A. Drift Parameters

Examples:

- $\alpha_F, \beta_F, \kappa_F$
- $\alpha_S, \gamma_F, \gamma_D, \beta_S$
- $\alpha_R, \eta_F, \beta_R, \kappa_R$
- $\alpha_D, \eta_D, \beta_D$

These govern continuous evolution rates.

B. Diffusion Parameters

$$\sigma_F > \sigma_S > \sigma_R > \sigma_D$$

Constrained ordering enforced during estimation.

C. Jump Intensity Parameters

$$\lambda_0, \lambda_F, \lambda_S, \lambda_u, \lambda_Z$$

Separate irreducible vs modifiable risk components.

D. Jump Magnitude Parameters

ϕ_1, ϕ_2,variance parameters

Determine structural loss severity.

E. Performance Parameters

$$a_F, a_D, b_R, c_S, \sigma_P$$

Control mean and variance structure.

F. Regime Threshold Parameters

Define geometry of:

$$\mathcal{R}_{minor}, \mathcal{R}_{major}, \mathcal{R}_{cat}$$

These are interpretable medical thresholds.

G. Non-Negotiable Parameter Constraints (Freeze Policy)

Certain parameters are fixed or globally pooled:

- Irreducible jump baseline λ_0 is league-level constant.
- Diffusion ordering enforced by constrained optimisation.
- Fatigue mean-reversion bounded within fixed interval.
- Catastrophic threshold anchored to historical incidence.

Catastrophic Incidence Calibration.
The catastrophic injury boundary parameter(s) are not freely estimated from sparse player-level data. Instead, they are calibrated to reproduce empirical season-level catastrophic injury incidence under baseline workload conditions.

Let π_{hist} denote the historical catastrophic injury rate per season (league-level or competition-level).

Let $\Psi_\theta(T)$ denote the model-implied season survival probability under baseline workload and parameter set θ.

The catastrophic threshold parameter(s) are chosen such that:

$$1 - \Psi_\theta(T) = \pi_{hist}$$

under baseline policy.

Baseline workload policy is defined as empirically observed average workload under historical conditions.

Calibration is performed prior to player-specific parameter estimation to ensure global consistency.

This calibration ensures that catastrophic injury frequency is anchored to empirical incidence rather than inferred from sparse tail events.

Robustness Check.
Sensitivity analysis is performed by varying π_{hist}within confidence bounds derived from historical data to evaluate stability of catastrophic boundary placement.

10.2 Hierarchical Structure

To prevent overfitting and improve identifiability:

Parameters follow hierarchical model:

$$\theta_i \sim \mathcal{N}(\mu_\theta, \Sigma_\theta)$$

Where:

- $i$indexes players.
- Hyperparameters estimated across team or league.
- Allows partial pooling.

This ensures:

- Data-poor players borrow strength.
- Overfitting is controlled.
- Structural coherence maintained.

10.3 Identifiability Enforcement

Structural restrictions include:

1. Time-scale separation (drift magnitude priors).
2. Diffusion ordering constraints.
3. Restricted cross-interaction graph.
4. Bounded curvature assumptions.
5. Non-overlapping measurement loading matrix.

We do NOT allow fully flexible nonlinear interactions.

This preserves interpretability.

10.4 Estimation Methodology

Because:

- State is latent
- Dynamics nonlinear
- Jump process present
- Observations non-Gaussian

Estimation proceeds via:

Particle MCMC or Sequential Monte Carlo EM:

1. Particle filter for state estimation.
2. Likelihood approximation via Monte Carlo.
3. Parameter update via:
    - Stochastic gradient ascent, or
    - EM-style expectation maximisation.

10.5 Data Requirements

Model requires:

- Time-stamped workload data
- Injury occurrence data
- Physiological proxy data
- Match performance data
- Recovery input data

Longitudinal structure is essential.

Cross-sectional data insufficient.

10.6 Calibration Phases

Estimation proceeds in staged fashion:

Phase 1:

- Calibrate fatigue subsystem independently (fast scale).

Phase 2:

- Calibrate structural and micro-damage drift.

Phase 3:

- Estimate jump intensity using injury events.

Phase 4:

- Estimate performance mapping.

This staged approach improves identifiability.

10.7 Validation

Model validated via:

- Out-of-sample injury timing prediction.
- Calibration of survival curves.
- Brier score for injury probability.
- Performance forecast error.
- Stress-test robustness checks.

We do not rely solely on predictive accuracy.

We test structural plausibility.

10.8 Computational Considerations

Computation involves:

- Parallel particle filtering
- Monte Carlo policy simulation
- Sensitivity analysis
- Bootstrapped uncertainty estimation

Model designed to be computationally heavy but feasible.

Section 11 – Structural Assumptions, Limitations & Ethical Considerations

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11.1 Structural Assumptions

The model rests on the following explicit structural assumptions:

A. Low-Dimensional Latent Health

Health is represented by a 4-dimensional latent vector:

$$\mathbf{H}_t = (S_t, F_t, R_t, D_t)$$

This is an abstraction.

True physiological complexity is higher dimensional.

The model assumes that these four components capture dominant dynamics relevant to injury and performance.

B. Time-Scale Separation

Fatigue evolves faster than structure and micro-damage.

Structural degradation evolves faster than cumulative micro-damage.

This separation is imposed to preserve identifiability and interpretability.

It is an approximation, not a biological certainty.

C. Jump Process Representation

Acute injury shocks are modeled as Poisson jump events.

This assumes:

- Memoryless arrival conditional on state.
- Independent increments.

Real-world injury processes may have clustering effects not fully captured.

D. Region-Based Injury Definition

Injury is defined as first entry into predefined regions of health space.

Threshold geometry is assumed stable over season.

True injury mechanisms may involve more complex biomechanical triggers.

E. Piecewise-Constant Control

Workload is assumed constant within decision intervals.

This approximates real planning blocks but ignores micro-adjustments during games.

F. Approximate Control

The model does not solve exact belief-state dynamic programming.

Instead, it uses:

- Particle filtering
- Monte Carlo simulation
- Approximate policy optimisation

Optimality is approximate.

11.2 Estimation Limitations

- Particle filtering may suffer from degeneracy under sparse data.
- Parameter estimation may be weakly identified in short seasons.
- Injury events are rare, making jump intensity estimation difficult.
- Measurement error may bias structural inference.

These risks are mitigated through hierarchical priors and staged calibration.

11.3 Computational Limitations

- Monte Carlo simulation may be computationally intensive.
- Policy optimisation may require significant parallel computation.
- Real-time implementation requires engineering support.

11.4 Model Risk

This model is itself subject to model risk:

- Misspecified drift dynamics
- Incorrect threshold geometry
- Underestimated jump intensity
- Overconfident filtering

To mitigate:

- Regular backtesting
- Stress testing
- Sensitivity analysis
- Conservative medical constraint enforcement

11.5 Ethical Considerations

The system must not:

- Encourage overexploitation of athletes.
- Justify pushing players to catastrophic boundary.
- Override medical authority.

Medical constraint is binding.

Model supports, not replaces, medical judgment.

Player welfare remains primary.

11.6 Interpretational Boundaries

Outputs should be interpreted as:

- Probabilistic risk estimates.
- Decision-support signals.
- Not deterministic injury predictions.

Uncertainty intervals must accompany all reported metrics.

11.7 Extensibility

The framework can be extended to:

- Multi-player team aggregation
- Correlated shock modeling
- Tournament-level scheduling stress testing
- Career-level sustainability modeling

But these are outside current scope.

Section 12 — Structural Stability Safeguards & Rare-Event Governance

12.1 Filtering Stability & Degeneracy Control

Because the model combines:

- Jump processes
- Slow-moving latent states
- Low diffusion components

- Rare catastrophic transitions

Particle degeneracy is a structural risk.

To prevent silent collapse:

12.1.1 Diffusion Floor Enforcement

All state components satisfy:

$$\sigma_i \geq \sigma_{\min} > 0$$

including slow states S_t, D_t.

This ensures:

- Persistent particle dispersion.
- Prevention of deterministic collapse.
- Honest uncertainty propagation.

12.1.2 Systematic Rejuvenation

At every decision epoch t_k:

- An MCMC rejuvenation kernel is applied to particle cloud.
- Kernel targets slow states explicitly.
- Rejuvenation step size calibrated to preserve posterior structure.

Rejuvenation is not conditional — it is mandatory.

12.1.3 Effective Sample Size (ESS) Monitoring

At each update:

$$ESS_k = \frac{1}{\sum w_i^2}$$

If $ESS_k < \tau_{ESS}$, additional rejuvenation is triggered.

This prevents silent weight collapse.

12.1.4 Likelihood Tempering

If observation likelihood produces extreme weight concentration:

- Tempered likelihood updates are applied.

- Sequential importance rescaling used.

This prevents single observation shock from collapsing belief.

12.1.5 Filtering Stress Testing

Before deployment:

- Simulated ground-truth experiments performed.
- Recovery of slow states evaluated.
- Catastrophic tail probability recovery tested.

Filtering engine must pass synthetic identifiability tests before real data use.

12.2 Hierarchical Injury Region Anchoring

Injury regions must not be arbitrarily estimated.

To prevent geometric overfitting:

12.2.1 Nested Region Constraint

Regions satisfy strict nesting:

$$\mathcal{R}_{cat} \subset \mathcal{R}_{major} \subset \mathcal{R}_{minor}$$

This enforces structural hierarchy.

12.2.2 Structural Integrity Anchoring

Catastrophic region must include structural threshold:

$$S_t \leq S_{cat}$$

or joint structural form:

$$S_t + \omega D_t \leq C_{cat}$$

Catastrophic injury cannot occur without structural collapse component.

12.2.3 Low-Dimensional Parametric Geometry

Region geometry restrictions follow Section 5.3 and are enforced during calibration.

12.2.4 Medical Calibration Anchors

Threshold parameters are anchored using:

- Historical catastrophic injury rates.
- Clinical injury severity definitions.
- External medical literature priors.

Regions are not purely data-driven.

12.3 Rare-Event Parameter Governance

Catastrophic injuries are rare.

Unconstrained estimation leads to instability.

Therefore:

12.3.1 Fixed Irreducible Jump Baseline

$$\lambda_0 = \lambda_{league}$$

Shared across players.

Not individually estimated unless large data available.

12.3.2 Constrained Jump Sensitivity Structure

Sensitivity parameters share functional form across players.

Heterogeneity allowed only via scaling factors.

Sensitivity parameters are subject to shrinkage regularisation and may be partially pooled toward baseline if empirical injury frequency is insufficient to support independent estimation.

12.3.3 Catastrophic Threshold Anchoring

Catastrophic boundary calibrated to reproduce:

- Empirical season-level catastrophic incidence.

Threshold is not freely estimated from sparse player data.

12.3.4 Drift Curvature Bounds

Fatigue mean-reversion and structural repair parameters are bounded within physiologically plausible ranges.

This prevents drift absorbing rare-event effects.

12.4 Identifiability Stress Testing Protocol

Before model approval:

1. Simulate synthetic seasons with known parameters.
2. Fit model to synthetic data.
3. Evaluate:
   - Recovery of slow-state trajectories.
   - Jump intensity estimation accuracy.
   - Catastrophic probability calibration.
4. Evaluate sensitivity to:
   - Reduced observation quality.
   - Missing recovery inputs.
   - Sparse injury events.

Model must demonstrate stable inference under stress.

Identifiability stress tests include posterior recovery of region boundary parameters.

12.5 Conservative Risk Reporting

All catastrophic probability outputs must include:

- Monte Carlo uncertainty interval.
- Sensitivity band under plausible parameter variation.
- Filter uncertainty dispersion.

No single-point catastrophic probability may be reported without uncertainty bounds.

12.6 Governance Override Principle

If:

- Posterior dispersion is high, or
- ESS instability persists, or
- Rare-event calibration uncertainty exceeds threshold,

Then:

- Medical constraint is tightened conservatively.

- Aggressive workload policies are disabled.

Model uncertainty triggers conservatism automatically.

12.7 Structural Rigidity Commitment

The model explicitly prioritises:

- Stability
- Identifiability
- Governance credibility

over:

- Parametric flexibility
- Maximum predictive fit
- Complex free-form geometry

Structural rigidity is intentional.